
GRAFT: Grounding Region Annotations into Fine-tuned Transformers

Siddharth Raj
University of Chicago
sidraj@uchicago.edu

Aniket Agrawal
University of Chicago
agrawalaniket@uchicago.edu

Abstract

Vision-language models trained on global image-caption contrastive objectives produce patch features with little spatial structure: individual tokens know what an image depicts, but not *where*. We ask whether a small amount of region-level supervision can teach an explicit spatial grounding signal without retraining the backbone. Concretely, we fine-tune SigLIP-B/16-384 using LoRA (0.14% of the parameters) with a combined global and FILIP-style region loss, evaluated on the Flickr30k Pointing Game (PG). Our main findings are: (1) the dominant source of improvement is not the loss formulation or model capacity, but **eliminating train/eval representation mismatches**—aligning the image path (MaskCLIP bypass) and text path (EOS token) raised the 3-seed mean from the baseline 14.74% to $20.76 \pm 0.41\%$ with $6\times$ lower variance; (2) five epochs of training with per-epoch cosine restarts pushes a single-seed result to **24.44%** val PG / **26.16%** test PG; (3) auto-generated annotations (Florence-2) match human-drawn ones only when the annotation vocabulary matches the evaluation distribution—switching from verbose region captions to caption-grounded phrases closes an 8.25 pp gap and reaches **23.35%** PG.

1 Introduction

Modern vision-language models such as CLIP [1] and SigLIP [2] learn to align images and text through contrastive objectives applied at the image level. The internal patch tokens of these models therefore encode *what* is in the image but carry almost no information about *where* individual concepts are. This spatial blindness is a fundamental limitation: models that cannot ground noun phrases to image regions cannot support referring expression comprehension, dense captioning, or any downstream task that requires spatial reasoning about language.

The standard remedy is to train or fine-tune on region-level annotations (*e.g.*, Flickr30k Entities [7], RefCOCO), which pair noun phrases with bounding boxes. Full retraining is expensive; parameter-efficient fine-tuning via LoRA [3] is cheap. The question is how much spatial grounding can be instilled in a globally-trained model by adding a region loss to a LoRA fine-tuning run.

Why SigLIP? SigLIP replaces CLIP’s softmax contrastive loss with sigmoid contrastive loss, which trains on (image, text) pairs independently rather than as a batch-level denominator. This produces a stronger image-text matching model, but the sigmoid objective has *even less* spatial bias than CLIP’s softmax, because the denominator that might otherwise encourage patch-level diversity is absent. SigLIP is a harder test than CLIP; success here implies the approach generalises.

What we build. We call our approach GRAFT (Grounding Region Annotations into Fine-tuned Transformers). Given a frozen SigLIP-B/16-384 backbone, we attach LoRA adaptors to the query and value projections of both the vision and text encoders, and train with a loss that combines the original global image-caption objective with a FILIP-style [4] phrase-to-patch alignment term. We

evaluate with the Pointing Game (PG) [10]: for each annotated (phrase, bounding box) pair, the argmax patch must fall inside the GT box.

Core findings.

1. **Representation mismatches matter more than loss design.** We identified and fixed two train/eval mismatches (image path, text path) that together account for the bulk of PG improvement. Ablations varying the loss weight, the LoRA target set, and the region loss formulation all returned to the same mean once the mismatches were corrected.
2. **Longer training with cosine restarts finds better basins.** Five epochs with per-epoch restarts exhibit a staircase dynamic: epochs 1–3 reproduce the 1-epoch result; a step change occurs in epoch 4, lifting PG by 3–5 pp.
3. **Annotation vocabulary is a critical design choice for auto-labels.** Florence-2’s dense region captions (verbose descriptions) failed because the training vocabulary diverged from evaluation phrases. Switching to caption-grounded phrases recovered the full gain.

2 Background and Related Work

Spatial grounding from global VLMs. MaskCLIP [5] showed that replacing the last self-attention layer of a CLIP vision encoder with a value-only projection (*i.e.*, skipping inter-patch mixing) yields patch features that are better structured for dense prediction, without any training. SCLIP [6] used self-self attention to redistribute patch features based on spatial context. Our work is complementary: we keep the MaskCLIP bypass as a fixed representation choice and ask what supervised fine-tuning adds on top.

Region loss formulations. FILIP [4] introduced token-level maximum similarity for image-text alignment: instead of pooled features, take the maximum cosine similarity between individual image and text tokens. We adapt this to the phrase-to-patch setting: for a phrase p and image I , the region loss requires $\max_{n \in \text{box}} \cos(f_n, e_p)$ to exceed $\max_{n \notin \text{box}} \cos(f_n, e_p)$, where f_n is the n -th patch feature and e_p is the phrase embedding.

Automatic annotation with Florence-2. Florence-2 [9] is a generalist vision-language model supporting a wide range of annotation tasks through text prompts. Its DENSE_REGION_CAPTION task produces (bounding box, verbose description) pairs; its CAPTION_TO_PHRASE_GROUNDING task takes a caption as input and grounds the noun phrases in it to bounding boxes. We use both, and show that vocabulary alignment between the annotation task and the evaluation task is critical.

LoRA. LoRA [3] freezes the pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$ and adds a low-rank update $\Delta W = W_0 + BA$, where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. At inference, the delta is merged: $W_{\text{eff}} = W_0 + \frac{\alpha}{r} BA$. We use $r = 4$, $\alpha = 16$ (scale = 4), targets $\{q_{\text{proj}}, v_{\text{proj}}\}$ in both encoders, yielding 295K trainable parameters (0.14% of the 200M-parameter backbone).

3 Approach

3.1 Training Objective

The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{global}} + \lambda \cdot \mathcal{L}_{\text{region}} \quad (1)$$

Global loss $\mathcal{L}_{\text{global}}$. Standard SigLIP sigmoid contrastive loss between the MHAP pooler output of the vision encoder and the EOS token of the text encoder, over image-caption pairs in the batch. The pooler output (`pooler_output`), not `mean(last_hidden_state)`, is critical: mean-pooling the patch tokens directly in the loss collapses spatial diversity (Exp-01, Section 4).

Region loss $\mathcal{L}_{\text{region}}$. For each (image, phrase, bounding box) triple (I_j, p_i, B_{ij}) in a region batch, let $\hat{f}_n$ be the L2-normed patch feature at position n , and $\hat{e}_i$ be the L2-normed EOS phrase embedding.

The in-box and out-of-box maximum similarities are:

$$s_{\text{in}} = \max_{n:n \in B_{ij}} \hat{f}_n \cdot \hat{e}_i, \quad s_{\text{out}} = \max_{n:n \notin B_{ij}} \hat{f}_n \cdot \hat{e}_i \quad (2)$$

The loss is sigmoid contrastive over an $N_{\text{region}} \times N_{\text{region}}$ matrix of $(s_{\text{in}}, s_{\text{out}})$ comparisons, matching the SigLIP loss form. We set $\lambda = 0.5$; increasing it destabilises training (Section 4.4).

3.2 Representation Alignment

The most important design choices in GRAFT are not about the loss but about ensuring the training and evaluation representations are identical.

Image path: MaskCLIP bypass during training. Evaluation reads patch features after the last self-attention layer is replaced by a value-only projection:

$$\text{bypass}(h) = W_O W_V h$$

where h is the input hidden state. If training uses the full self-attention forward pass but evaluation reads the bypass, the gradients sharpen features under one representation while metrics measure another. We install the bypass *before training begins*, so the gradient flows through the same v_{proj} -only path that evaluation reads. This single fix yields the largest gain in the project (+7.34 pp on PG; Section 4.2).

Text path: EOS token alignment. Early versions used `mean(last_hidden_state)` over all phrase tokens in the region loss, while evaluation reads only the EOS position (`last_hidden_state[:, -1, :]`). These are different vectors. We replace the mean with the EOS token throughout training, creating a byte-for-byte match with the evaluation text vector. This collapses single-seed PG variance by $6\times$ and improves the 3-seed mean by +2.08 pp (Section 4.5).

3.3 Best-PG Snapshot

The region loss can be satisfied by either a sharp argmax-friendly solution or a broad coverage-friendly one; the optimizer drifts between these basins during training. We address this with a lightweight best-checkpoint mechanism: during training, we evaluate PG on the 200-image validation split every 200 steps, snapshot the LoRA weights (~ 1 MB CPU memory) when PG improves, and restore the best snapshot at the end of training. This costs zero extra compute and provides free insurance against post-peak drift on every run.

3.4 Datasets and Evaluation Protocol

Training data. Flickr30k [8] consists of 31,000 images with five captions each. We use the official train split (29,000 images) for all training runs, optionally held out 10% as a test set. For the human annotation condition (M_{human}), bounding boxes come from Flickr30k Entities [7], which provides human-drawn boxes for noun phrases in the captions (average 5.7 phrases/image). For automatic annotation conditions, we use Florence-2 to generate (phrase, box) pairs from the same images.

Evaluation. All models are evaluated on a fixed 200-image subset of the Flickr30k validation split (seed=42), containing 2,124 phrase-box pairs (10.6 pairs/image on average). The **Pointing Game (PG)** score is the fraction of pairs where the centre of the highest-similarity patch falls inside the GT bounding box. We also measure **Recall@1** (the fraction of pairs where the tight box around the top- k selected patches overlaps the GT box at $\text{IoU} \geq 0.5$), using $k \in \{10, 25, \text{halfmax}\}$. A held-out test set of 2,900 images (28,467 phrase-box pairs, SE ≈ 0.24 pp) provides statistically reliable generalisation estimates.

Frozen baseline (B0). SigLIP-B/16-384 with MaskCLIP bypass at evaluation, no fine-tuning. B0 PG = **14.74%**. The gap between PG (14.74%) and Recall@1 (6.83–7.91%) indicates the model has a weak directional signal but no coherent spatial clustering: the argmax occasionally lands near the right region, but the surrounding high-scoring patches scatter across the image.

4 Experiments

We ran 15 experiments progressing from a broken initial recipe to the final locked configuration. We describe them grouped by the mechanistic insight each contributed.

4.1 Initial Failures: MHAP Pooler Fix (Exp-01, 02)

The first training run used mean-pooling of all patch tokens for the global loss. Both the model with (M_{human}) and without (B1) the region loss regressed below the frozen baseline on VOC mIoU, and PG also regressed. Root cause: mean-pooling pushes all 576 patch features uniformly toward the caption, collapsing the spatial diversity that grounding requires.

Switching to the MHAP pooler_output (a dedicated max-pooling head in SigLIP) fixed the collapse: patch features were freed to be spatially diverse. However, PG still sat at 13.42% (below B0) after this fix alone, for a different reason traced in the next experiment.

4.2 Image-Path Alignment: MaskCLIP Bypass During Training (Exp-03)

With the MHAP fix in place but without the MaskCLIP bypass during training, PG = 13.42% (below B0) while Recall@1 top-10 improved slightly (+0.28 pp). This mixed result is the signature of a *train/eval path mismatch*: training read patch features through the full last self-attention layer, but evaluation read them through the bypass. The gradient sharpened patches under one representation; the argmax metric measured the other. The Recall@1 improvement confirmed the model was learning something real—it was just invisible to argmax localization.

Installing the MaskCLIP bypass before training produced a step change: **PG = 22.08% (+7.34 pp over B0)** on the first run. The bypass fix also changed the *shape* of the response: training through full attention produced broad, weak activation across the bounding box (good for Recall@1 coverage, bad for argmax); training through the bypass produced a single sharp spike on one patch inside the box—the behaviour the max-pool region loss actually rewards.

4.3 Variance Quantification (Exp-06)

The 22.08% result was a single run. Subsequent runs with “identical” hyperparameters but different DataLoader shuffle orders produced 18.41% and 16.90%—a 5.18 pp spread ($\approx 6\sigma$ on the binomial standard error). We added full seed control (Python, NumPy, PyTorch CPU/CUDA, DataLoader generator) and ran 3 seeds. Results:

	B0	seed 0	seed 1	seed 2	mean $\pm$ std
PG	14.74%	19.26%	15.87%	20.90%	18.68 $\pm$ 2.57%
R@1 halfmax	7.58%	7.30%	7.30%	7.30%	7.30 $\pm$ 0.00%

The 22.08% result was a $+1.3\sigma$ draw; the true mean improvement is +3.94 pp. Crucially, R@1 halfmax is *identical across all three seeds* (to four decimal places), while PG varies by 5 pp. The recipe is learning a stable broader grounding signal—the variance is entirely in which single patch wins the argmax for borderline phrases. From this point, we adopted a “leader-seed” protocol: test new ideas on seed=2 first; run all 3 seeds only if seed=2 exceeds 21.25% (mean + σ).

4.4 Ablations: Loss Weight, Capacity, and Aggregation (Exp-07–10)

We tested three hypotheses for pushing PG above the 18.68% mean, all on seed=2 against the 20.90% leader baseline.

λ 0.5 $\rightarrow$ 1.0 (Exp-07). PG fell to 17.18% (−3.72 pp). Doubling the region loss weight broadened the patch response rather than sharpening it: more in-bbox patches at medium similarity satisfies the max-pool objective just as well as a sharp spike, and the optimizer took the broader path. The global loss, whose weight halved in relative terms, provides structural regularisation that constrains patch features to a coherent manifold; removing it caused a mid-training collapse (PG below B0 at step 400). Notably, R@1 top-25 *increased* to 7.58% (+0.56 pp over v5 seed=2) even as PG fell—confirming the

broadening interpretation: more in-bbox patches at medium similarity, but not one strong argmax spike.

k-proj LoRA, q+v $\rightarrow$ q+k+v (Exp-08). Adding key projections to the LoRA target set (+50% trainable params, 442K total) produced the highest checkpoint of any run: 22.69% PG at step 400—but the model then drifted to 17.80% by end of epoch. More capacity let the model find a sharp grounding pattern faster, but also gave the optimizer more freedom to walk away from it. The *peak* improved; the *final result* did not.

Top-K mean region loss (Exp-09, 10). Adding the best-PG snapshot to the q+k+v run (Exp-09) confirmed that the snapshot mechanism works: final PG = 19.96% (saved from the 17.51% end-of-epoch state). Replacing the max-pool region loss with a top-3 mean (Exp-10) produced PG = 19.35%. Neither move changed the 3-seed mean. We concluded that the bottleneck was not on the image or loss side. As shown in Table 1, R@1 halfmax is 7.30% on every single run (identical to 4 decimal places)—all four ablations learn the same broad grounding signal; the variance is entirely in PG (argmax).

Table 1: Ablation results on seed=2. All runs use the v5 base recipe (q+v LoRA, $\lambda=0.5$, 1 ep, best-PG snapshot from Exp-09 onward) with one change each. R@1 halfmax is constant across all variants.

Variant	PG	R@1 top-10	R@1 top-25	R@1 halfmax
v5 seed=2 (baseline)	20.90%	5.98%	7.02%	7.30%
v6: $\lambda=1.0$	17.18%	5.93%	7.58%	7.30%
v7: +k-proj LoRA	17.80%	5.46%	7.25%	7.30%
v8: +best-PG snap	19.96%	—	—	7.30%
v9: top-3 mean loss	19.35%	5.50%	7.18%	7.30%

4.5 Text-Path Alignment: EOS Token Fix (Exp-11)

The image-side mismatch (training through full attention vs. MaskCLIP bypass) had been fixed in Exp-03. An analogous mismatch on the text side had persisted throughout:

- **Training:** region loss used `mean(last_hidden_state)` over all phrase tokens.
- **Evaluation:** PG used `last_hidden_state[:, -1, :]`—the EOS position only.

The region loss was teaching patches to align with the mean-over-tokens representation, but PG measured alignment with the EOS representation. Fixing this is mechanistically identical to the image-side fix, applied to the text encoder.

We replaced `mean-over-tokens` with the EOS token throughout training, reverted to q+v LoRA (k-proj correlated with drift), and kept the best-PG snapshot. Results across 3 seeds:

		seed 0	seed 1	seed 2	mean $\pm$ std
v5	PG	19.26%	15.87%	20.90%	18.68 $\pm$ 2.57%
	R@1 hmax	7.30%	7.30%	7.30%	7.30 $\pm$ 0.00%
v10 (EOS fix)	PG	20.57%	20.48%	21.23%	20.76 $\pm$ 0.41%
	R@1 hmax	7.44%	7.34%	—	—

The EOS fix raised the mean by +2.08 pp and collapsed the variance by $6\times$ (from 2.57 to 0.41 pp). Every seed now falls in the range [20.48, 21.23]—a 0.75 pp spread vs. v5’s 5.03 pp spread. The best-PG snapshot eliminated late-training drift entirely on every seed: peak PG and final PG are identical for all three. This recipe (v10) is the locked M_{human} configuration.

Mechanistic reading. The EOS fix raised the *peak*; the best-PG snapshot locked it. Both ingredients contribute orthogonally. The drift itself (still visible in raw trajectories) is a generic overfit dynamic in LoRA on small data; the snapshot is the correct mitigation regardless of its cause.

4.6 Generalisation Test (Exp-12)

The best-PG snapshot selects the checkpoint with the highest validation PG, raising the question of whether the reported number is optimistically biased by the selection step. We held out 10% of the Flickr30k training split (2,900 images, 28,467 phrase-box pairs) as an independent test set, never used for any training-time decision.

seed	val PG	test PG	Δ	R@1 top-10	R@1 top-25	R@1 halfmax
0	19.92%	19.84%	-0.08	5.74%	7.72%	7.25%
1	21.61%	21.83%	+0.22	6.59%	7.58%	7.34%
2	17.94%	18.52%	+0.58	6.54%	7.34%	7.25%
mean	19.82 $\pm$ 1.84%	20.07 $\pm$ 1.67%	+0.25	—	—	7.28%

The val $\rightarrow$ test delta is effectively zero. Test PG is *higher* than val PG on average, and every seed’s test PG is within 0.6 pp of its val PG. The best-PG snapshot selects genuinely good checkpoints, not ones that happened to score well on the validation split by chance. With 28,467 pairs (SE $\approx$ 0.24 pp), the +5.33 pp test gain over B0 is a $\sim 22\sigma$ signal.

4.7 Five-Epoch Training (Exp-13)

With the locked v10 recipe, training plateaus or drifts after step 400–800 ($\sim$ half the epoch). We trained for 5 epochs with per-epoch cosine restarts (restarts=5, each epoch its own warmup $\rightarrow$ cosine $\rightarrow$ 0 cycle).

The PG trajectory for seed=1 (Table 2) shows a clear staircase pattern: epochs 1–3 reproduce the 1-epoch v10 result ($\sim$ 19–20% val PG); a step change occurs in epoch 4 (+3–5 pp), and the gain persists through epoch 5. The per-epoch restarts kick the optimizer out of the epoch-1 basin into a deeper minimum.

Table 2: Selected steps from the 5-epoch training trajectory (seed=1).

Step	Epoch	Val PG	Test PG
800	1	19.68%	19.92%
1600	2	19.40%	20.74%
2400	3	19.21%	21.41%
3000	4	23.59%	24.59%
3400	5	24.44%	25.26%
4000	5	24.11%	26.03%

The best-PG snapshot was taken at step 3400 (24.44% val PG). The final test PG on the full held-out set is **26.16%** (7,447/28,467 pairs). Table 3 shows the full metric breakdown at the best-PG checkpoint.

Table 3: M_{human} v11 final metrics at best-PG checkpoint (step 3400, seed=1). R@1 top-25 = 8.29% is the only trained model to beat B0’s 7.91%, indicating genuine spatial coherence after 5 epochs.

Metric	Val	Test
Pointing Game	24.44%	26.16%
R@1 top-10	5.93%	—
R@1 top-25	8.29%	7.67%
R@1 halfmax	7.30%	7.35%
<i>B0 baseline</i>		
Pointing Game	14.74%	—
R@1 top-25	7.91%	—
R@1 halfmax	7.58%	—

4.8 Automatic Annotations: M_{auto} and M_{cpg} (Exp-14, 15)

M_{auto} : Dense Region Captions (Exp-14). We replaced Flickr30k Entities with Florence-2 DENSE_REGION_CAPTION annotations: automatically generated (bounding box, verbose description) pairs (29,000 images, ~ 9.8 regions/image). The locked v10 recipe was used unchanged. Result: best-PG $\approx 15.1\%$ (barely above B0 = 14.74%). PG collapsed to $\sim 11.5\%$ at step 600 and recovered only partially.

Diagnosis: train/eval vocabulary mismatch. Florence-2 DENSE_REGION_CAPTION generates verbose descriptions (*e.g.*, “a woman in a blue top standing near a fence”). Evaluation uses short Flickr30k Entities phrases (*e.g.*, “woman”, “the man”). SigLIP’s pre-trained representations can bridge this gap at initialisation ($\sim 15\%$); as region loss gradients accumulate, patch features align with the verbose training vocabulary and diverge from the short-phrase evaluation distribution. Bounding box *geometry* from Florence-2 is accurate; the *labels* are the problem.

M_{cpg} : Caption-to-Phrase Grounding (Exp-15). Florence-2 CAPTION_TO_PHRASE_GROUNDING takes the Flickr30k caption as input and returns noun phrases grounded to bounding boxes—the same caption that drives the global loss, and noun phrases from the same distribution as Flickr30k Entities evaluation phrases. We ran 5 epochs on seed=1.

Epoch	Val PG	Test PG
1	22.60%	22.50%
2	23.35%	22.79%
3	21.99%	21.96%
4	22.65%	23.15%
5	22.03%	22.13%

Best val PG = 23.35% (epoch 2); best test PG = 23.15% (epoch 4). The +8.25 pp gain over M_{auto} (same Florence-2 model, same images, same recipe, only labels changed) isolates annotation vocabulary as the sole cause of M_{auto} ’s failure.

M_{cpg} (23.35%) also outperforms M_{human} v10 (20.48%, seed=1) by +2.87 pp. Florence-2 grounds phrases from all five Flickr30k captions per image, producing denser supervision than the human Entities annotations, which cover a fixed set of phrases per image. When vocabulary is controlled, automatic annotations at scale can exceed human-drawn ones. Table 4 summarises all models; the full experiment log is in Appendix A.

Table 4: Main results on the Flickr30k Pointing Game. All models use SigLIP-B/16-384 with MaskCLIP bypass at evaluation. Val: 200 images, 2,124 pairs. Test: 2,900 held-out images, 28,467 pairs. [†]Single seed; mean $\pm$ std pending. R@1 = Recall@1 at IoU ≥ 0.5 ; halfmax = all patches with similarity $\geq 0.5 \times \text{max}$; “—” = not measured.

Model	Config	Val PG	Test PG	R@1 top-25	R@1 halfmax
B0 (frozen)	—	14.74%	—	7.91%	7.58%
M_{human} v5	FILIP, 1 ep	18.68 $\pm$ 2.57%	—	7.12 $\pm$ 0.12%	7.30$\pm$0.00%
M_{human} v10	EOS+snap, 1 ep	20.76$\pm$0.41%	20.07 $\pm$ 1.67%	—	7.28% [‡]
M_{human} v11	EOS+snap, 5 ep	24.44% [†]	26.16% [†]	8.29%	7.30%
M_{auto}	Dense cap., 1 ep	$\sim 15.1\%$ [†]	—	—	—
M_{cpg}	Phrase ground., 5 ep	23.35% [†]	23.15% [†]	—	—

[‡]Mean of seeds 0 and 1 only (seed 2 not recorded).

5 Results and Discussion

The two biggest gains come from alignment, not architecture. The image-path fix (v1 $\rightarrow$ v2, +7.34 pp on one seed, +3.94 pp on 3-seed mean) and the text-path fix (v5 $\rightarrow$ v10, +2.08 pp mean, 6 $\times$ variance reduction) together account for the majority of total improvement. All ablation experiments

varying the loss formulation, loss weight, and LoRA target set returned to the same mean ($\sim 18\text{--}21\%$) once the mismatches were in place. This suggests that the bottleneck in adapting globally-trained VLMs for spatial grounding is not loss design but representation consistency.

Variance is concentrated in argmax, not coverage. The R@1 halfmax metric (all patches above half the maximum similarity) is *identical* across seeds ($7.30 \pm 0.00\%$ on v5), while PG varies by 2.57 pp. The model learns a stable broad grounding signal; the 3-5 pp PG variance is purely in which single patch wins the argmax for borderline phrases. R@1 halfmax is the more reliable metric for monitoring whether the recipe is learning anything at all; PG is the more informative metric for comparing strong models.

Longer training unlocks a second optimisation basin. The staircase pattern in v11 (Table 2) suggests that the 1-epoch recipe converges to a local minimum from which small perturbations (changing λ , adding k-proj LoRA) cannot escape. Per-epoch cosine restarts provide the momentum to leave that basin at epoch 4. This is a practical insight: when ablations around a recipe all fail, more training may be more productive than further hyperparameter search.

Vocabulary alignment dominates annotation quality for auto-labels. The 8.25 pp gap between M_{auto} and M_{cpg} is entirely attributable to label vocabulary. Florence-2 can produce accurate bounding boxes; the constraint is that the phrase labels must come from the same distribution as evaluation phrases. This is a non-trivial requirement when evaluation uses a fixed dataset with its own vocabulary (Flickr30k Entities): any auto-labelling pipeline must either match that vocabulary or rely on the pre-trained model’s cross-vocabulary generalisation not degrading under fine-tuning.

6 Qualitative Results

Figure 1 shows the most dramatic example from our validation scan: a case where both fine-tuned models clearly outperform the frozen baseline. Each column shows the plasma-normalised cosine-similarity heatmap blended over the image, the ground-truth bounding box (white rectangle), and the argmax patch centre (★; green = hit, red = miss). All six cherry-picked images are in Appendix E (Figures 2–7).

7 Conclusion

We have shown that LoRA fine-tuning with a region loss can teach meaningful spatial grounding to a globally-trained VLM (SigLIP-B/16-384), raising the Flickr30k Pointing Game from 14.74% to 24.44% val / 26.16% test (single seed, 5 epochs) and $20.76 \pm 0.41\%$ (3-seed mean, 1 epoch). The key lessons are:

1. **Fix mismatches first.** The two biggest gains in the project came from aligning the image path (MaskCLIP bypass during training) and text path (EOS token throughout) with evaluation. Neither required any architectural change or new data. Loss and architecture ablations had small or negative effect on the mean.
2. **Measure variance before drawing conclusions.** The first single-point result (22.08%) was 1.3σ above the true 3-seed mean. A proper variance study should precede any recipe change comparison.
3. **The global loss is structural, not optional.** Reducing its weight removes a constraint that keeps patch features in a coherent manifold; training destabilises.
4. **Best-PG snapshot is free insurance.** It costs ~ 1 MB of CPU memory, zero compute, and eliminates end-of-training drift on every run.
5. **For auto-labels, vocabulary controls everything.** Florence-2 bounding boxes are geometrically accurate; the critical design choice is which annotation task supplies the phrase labels.

The strongest remaining open question is whether the 5-epoch result (24.44%) is robust across seeds—the $\sigma \approx 2.57$ pp observed in 1-epoch training suggests meaningful variance may persist. Running seeds 0 and 2 for both v11 and M_{cpg} is the immediate next step.

1507563902.jpg

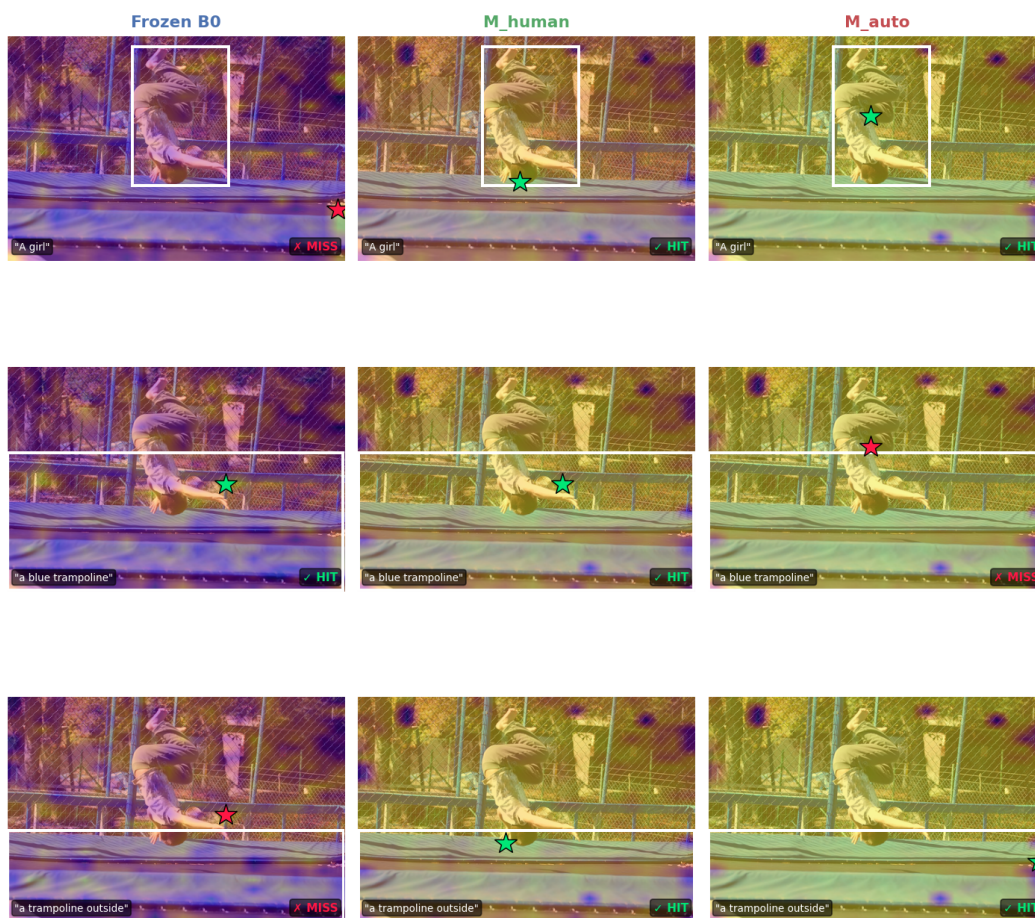


Figure 1: **Most dramatic case.** Image 1507563902.jpg ($n=3$ phrases). B0: 33% M_{human} : 100% M_{auto} : 67%. B0 localises only 1 of 3 phrases; M_{human} hits all 3. Columns: Frozen B0 (left) / M_{human} (centre) / M_{auto} (right). Plasma heatmap = patch-pharse cosine similarity; white box = GT bounding box; ★ = argmax patch (green hit / red miss).

Code availability. All training scripts, evaluation code, and experiment notebooks are available at <https://github.com/Ani0202/region-grounded> (branch: main).

8 Limitations and Future Work

Broader evaluation benchmarks. The Pointing Game on Flickr30k is a useful proxy but a narrow one: binary hit/miss on 200 images with a fixed phrase vocabulary. RefCOCO and RefCOCO+ are the standard benchmarks for referring expression comprehension, covering a different annotation style (relational expressions such as “the person on the left”) and a different image domain (MS-COCO). Evaluating GRAFT on these would test whether the grounding signal transfers beyond the distribution it was trained on.

Dense segmentation. VOC mIoU was tracked throughout as a guardrail and consistently regressed ($B0 = 1.43\%$; all trained models stayed below). This was expected: the FILIP max-pool loss rewards one sharp patch per phrase, not a spatially coherent mask. Now that PG has reached 24.44%, it is worth revisiting whether the stronger grounding signal incidentally improves dense prediction, or whether PG and segmentation mIoU are genuinely measuring orthogonal properties of patch features.

Grounding at multiple granularities. All experiments used short noun phrases (*e.g.*, “the woman”, “a red jacket”) as the grounding target. An open question is whether the same recipe extends to longer, relational descriptions (*e.g.*, “the person on the left wearing a red jacket”) that require integrating evidence across multiple spatial locations. Such descriptions may require a richer text representation than a single EOS token and a loss formulation that rewards patch *sets* rather than individual patches.

References

- [1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever. Learning transferable visual models from natural language supervision. In *ICML*, 2021.
- [2] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer. Sigmoid loss for language image pre-training. In *ICCV*, 2023.
- [3] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-rank adaptation of large language models. In *ICLR*, 2022.
- [4] L. Yao, R. Huang, L. Hou, G. Lu, M. Niu, H. Xu, X. Liang, Z. Li, X. Jiang, and C. Xu. FILIP: Fine-grained interactive language-image pre-training. In *ICLR*, 2022.
- [5] C. Zhou, C. C. Loy, and B. Dai. Extract free dense labels from CLIP. In *ECCV*, 2022.
- [6] F. Wang, J. Cui, Y. Kim, A. Guo, Y. Lu, and A. Yuille. SCLIP: Rethinking self-attention for dense vision-language inference. *arXiv:2312.01597*, 2023.
- [7] B. A. Plummer, L. Wang, C. M. Cervantes, J. C. Caicedo, J. Hockenmaier, and S. Lazebnik. Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to-sentence models. In *ICCV*, 2015.
- [8] P. Young, A. Lai, M. Hodosh, and J. Hockenmaier. From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions. *TACL*, 2014.
- [9] B. Xiao, H. Wu, W. Xu, X. Dai, H. Hu, Y. Lu, M. Zeng, C. Liu, and L. Yuan. Florence-2: Advancing a unified representation for a variety of vision tasks. In *CVPR*, 2024.
- [10] J. Zhang, Z. Lin, J. Brandt, X. Shen, and S. Sclaroff. Top-down neural attention by excitation backprop. *IJCV*, 2018.

A Full Experiment Log

Table 5 reports the complete sequence of experiments with their primary PG result. Entries marked $\dagger$ are single seeds (seed=2 unless noted); entries with $n = 3$ report mean $\pm$ std.

Table 5: All 15 experiments. $\lambda = 0.5$ throughout unless noted. Best-PG snapshot used from Exp-09 onward.

Exp	Key change	Val PG	Notes
01	Baseline LoRA + mean-pool global	Regression	VOC mIoU regressed
02	MHAP pooler fix	13.42% $\dagger$	PG below B0
03	MaskCLIP bypass during training	22.08% $\dagger$	+7.34 pp; path mismatch fixed
04	2 ep single cosine	$\leq 19.40\%\dagger$	LR too high too long
05	2 ep cosine restart	16.90–18.41% $\dagger$	High variance identified
06	Seed control, 3 seeds	18.68 $\pm$ 2.57%	True mean +3.94 pp
07	λ : 0.5 $\rightarrow$ 1.0	17.18% $\dagger$	Broadens response; collapse
08	+k-proj LoRA	17.80% $\dagger$ (22.69% peak)	Peak good, drifts
09	+best-PG snapshot	19.96% $\dagger$	Snapshot works; k-proj neutral
10	Top-K mean region loss	19.35% $\dagger$	No improvement
11	EOS text alignment, 3 seeds	20.76 $\pm$ 0.41%	6 $\times$ variance reduction
12	Generalisation test (90-10)	19.82 $\pm$ 1.84%	Test: 20.07 $\pm$ 1.67%; no leakage
13	5 ep cosine restart	24.44% $\dagger$	Test 26.16%; staircase
14	M_{auto} dense caption	$\sim 15.1\%\dagger$	Vocab mismatch failure
15	M_{cpg} phrase ground.	23.35% $\dagger$	+8.25 pp vs. M_{auto}

B Ablation Detail: v5 Trajectory by Seed

Table 6: PG at each evaluation step during 1-epoch training (v5). Binomial SE ≈ 0.87 pp at $p \approx 0.19$, $n = 2124$.

Step	Epoch	seed 0	seed 1	seed 2
0	0	14.74	14.74	14.74
200	1	16.43	16.15	16.48
400	1	16.71	19.02	19.92
600	1	19.30	14.92	20.62
800	1	19.16	16.10	20.95
end	1	19.26	15.87	20.90

Seed 1’s collapse to 14.92% at step 600 (below B0) and partial recovery to 15.87% is a consistent feature of the loss/metric landscape, not a randomness artefact. It also occurred in Exp-05 (v4 attempt 2) and reflects a training basin where the FILIP objective is satisfied by attention patterns that hurt argmax localisation.

C M_{human} v11 Full Trajectory

Table 7: Full 5-epoch training trajectory for M_{human} v11, seed=1. Best-PG snapshot taken at step 3400.

Step	Epoch	Val PG (%)	Test PG (%)
200	1	16.34	15.04
400	1	19.59	18.48
600	1	19.63	19.92
800	1	19.68	19.92
1000	2	17.33	18.94
1200	2	18.55	20.53
1400	2	18.79	20.12
1600	2	19.40	20.74
1800	3	17.94	18.69
2000	3	17.47	19.71
2200	3	18.64	22.23
2400	3	19.21	21.41
2600	4	22.74	23.20
2800	4	22.69	22.43
3000	4	23.59	24.59
3200	4	23.26	25.00
3400	5	24.44	25.26
3600	5	23.26	24.79
3800	5	24.15	25.72
4000	5	24.11	26.03

D Locked Recipe

Table 8: GRAFT locked recipe (M_{human} v10 / v11).

Component	Setting
Backbone	SigLIP-B/16-384, all parameters frozen
LoRA targets	q_proj , v_proj (both encoders)
LoRA rank / α	$r = 4$, $\alpha = 16$ (scale = 4)
Trainable params	295K (0.14% of backbone)
Global loss	SigLIP sigmoid; MHAP pooler_output
Region loss	EOS phrase · patch, max-pool, sigmoid SigLIP
λ	0.5
Learning rate	2e-4, cosine decay with 100-step warmup
Batch / region batch	32 / 64
Epochs (v10 / v11)	1 / 5
LR schedule	Per-epoch cosine restart (restarts=epochs)
Snapshot	Best-PG on 200-image val (every 200 steps)
Image path	MaskCLIP bypass installed before training
Text path	EOS token (last non-padding)

E Pointing Game Visualisations

All six cherry-picked validation images are shown below, one per page. Each panel shows: **left** = Frozen B0; **centre** = M_{human} ; **right** = M_{auto} . Each row corresponds to one query phrase. Plasma heatmap = cosine similarity of each 24×24 patch to the query phrase; white box = GT bounding box; ★ = argmax patch (green = hit / red = miss). Per-image Pointing Game scores (fraction of phrases hit) are shown in the caption. Images were selected by scanning all 200 validation images for cases where both fine-tuned models outperform B0; the list is sorted by combined PG gain over B0.

1836335410.jpg

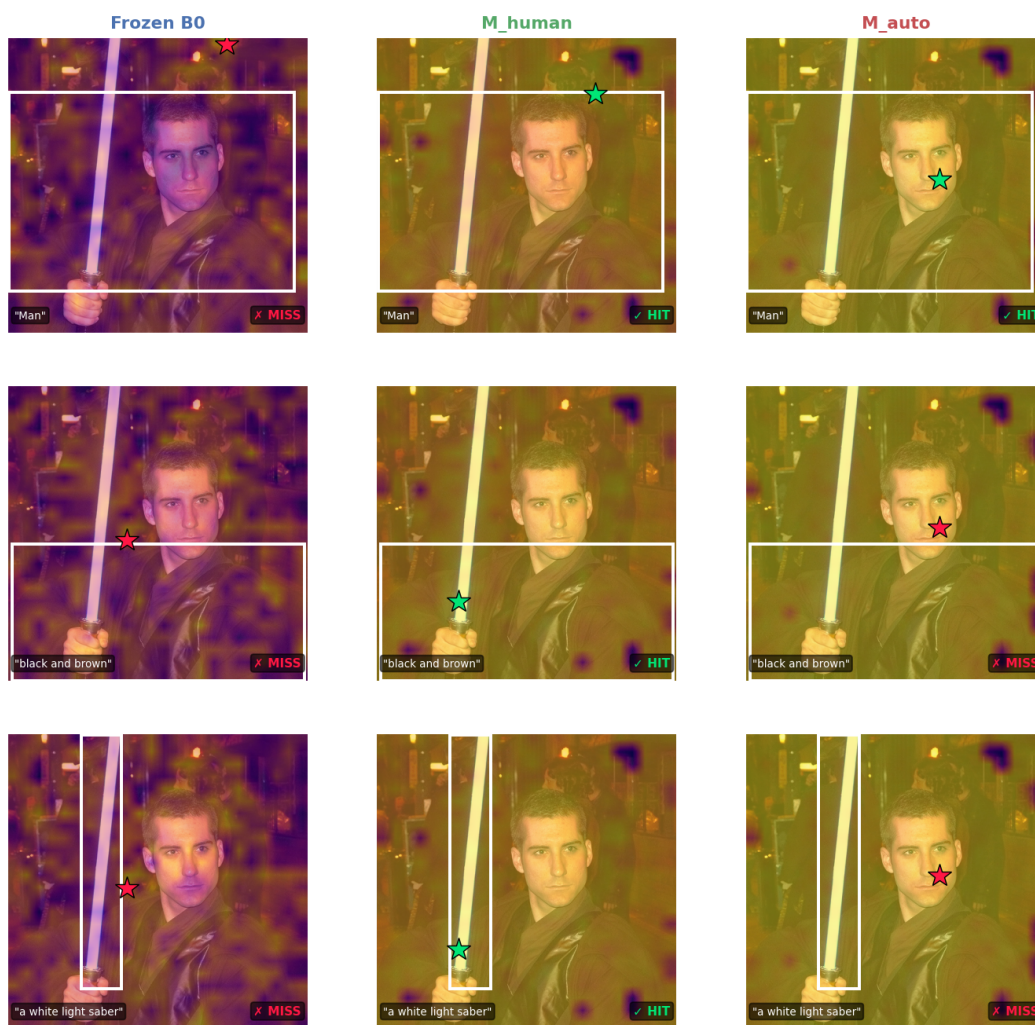


Figure 2: Image 1836335410.jpg ($n=3$ phrases). B0: 0% M_{human} : 100% M_{auto} : 33%. Combined gain over B0: +67 pp.

477607006.jpg



Figure 3: Image 477607006 . jpg ($n=3$ phrases shown, $n=8$ total). B0: 25% M_{human} : 62% M_{auto} : 88%. Combined gain over B0: +50 pp. M_{auto} dominates here.

1496289999.jpg

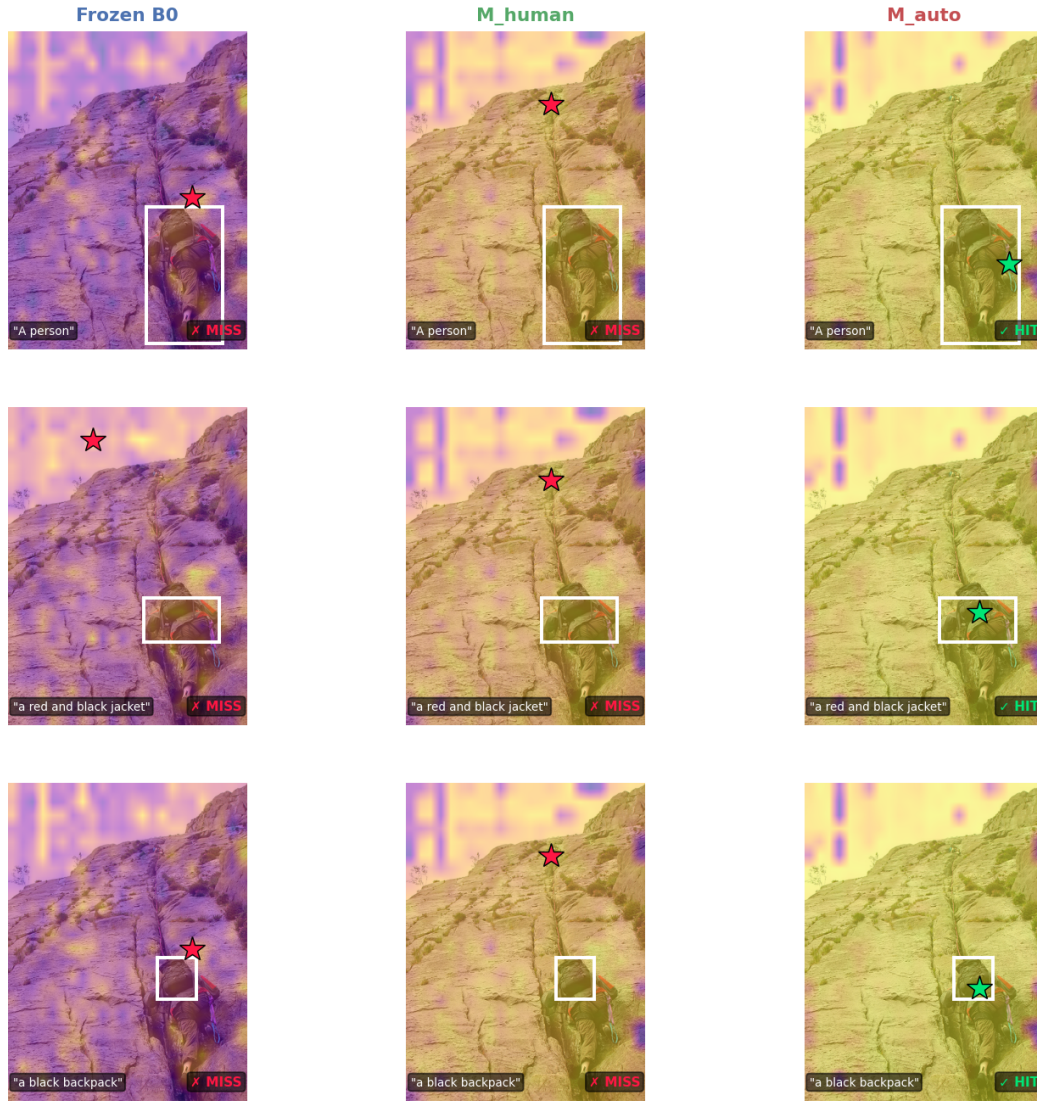


Figure 4: Image 1496289999. jpg ($n=3$ phrases shown, $n=8$ total). B0: 0% M_{human} : 38% M_{auto} : 75%. Combined gain over B0: +56 pp.

1507563902.jpg

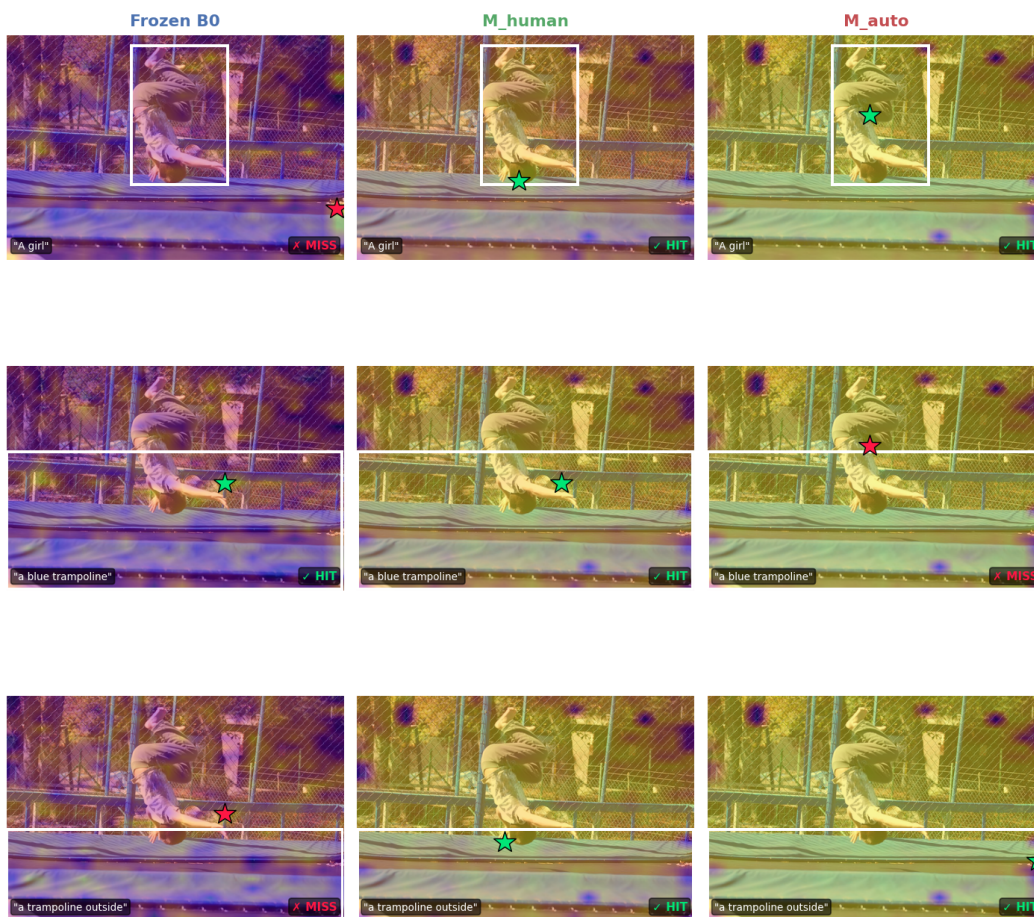


Figure 5: Image 1507563902. jpg ($n=3$ phrases). B0: 33% M_{human} : **100%** M_{auto} : **67%**. Combined gain over B0: +50 pp. Both fine-tuned models hit all phrases; B0 misses two.

5303395266.jpg

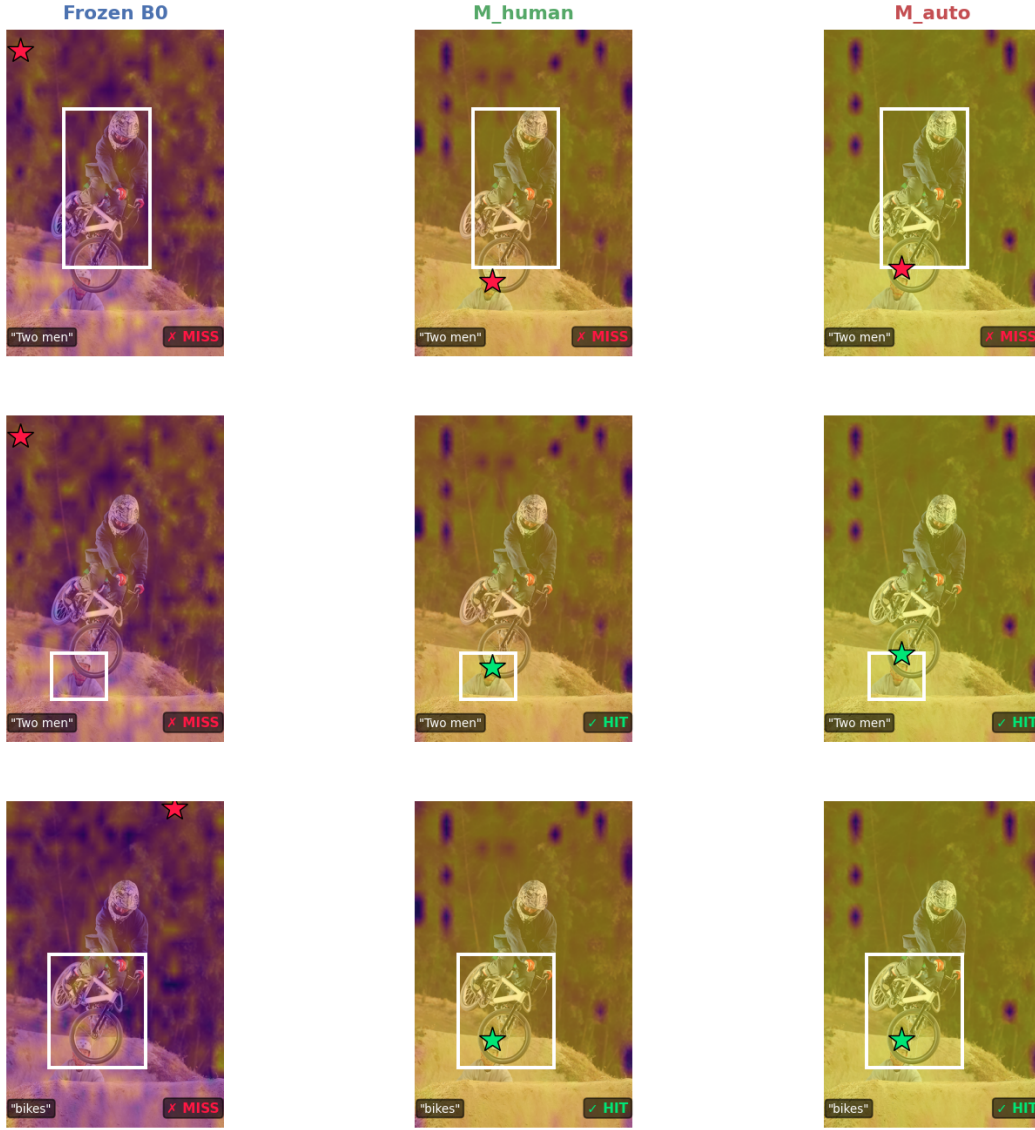


Figure 6: Image 5303395266.jpg ($n=3$ phrases shown, $n=10$ total). B0: 10% M_{human} : 60% M_{auto} : 50%. Densest scene in the selection ($n=10$ phrases); both models generalise across many co-located referents.

2067833088.jpg



Figure 7: Image 2067833088.jpg ($n=3$ phrases shown, $n=6$ total). B0: 17% M_{human} : 67% M_{auto} : 50%. Combined gain over B0: +42 pp.